

VectorFloorSeg: Two-Stream Graph Attention Network for Vectorized Roughcast Floorplan Segmentation

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Abstract

Vector graphics (VG) are ubiquitous in industrial designs. In this paper, we address semantic segmentation of a typical VG, i.e., roughcast floorplans with bare wall structures, whose output can be directly used for further applications like interior furnishing and room space modeling. Previous semantic segmentation works mostly process well-decorated floorplans in raster images and usually yield aliased boundaries and outlier fragments in segmented rooms, due to pixel-level segmentation that ignores the regular elements (e.g. line segments) in vector floorplans. To overcome these issues, we propose to fully utilize the regular elements in vector floorplans for more integral segmentation. Our pipeline predicts room segmentation from vector floorplans by dually classifying line segments as room boundaries, and regions partitioned by line segments as room segments. To fully exploit the structural relationships between lines and regions, we use two-stream graph neural networks to process the line segments and partitioned regions respectively, and devise a novel modulated graph attention layer to fuse the heterogeneous information from one stream to the other. Extensive experiments show that by directly operating on vector floorplans, we outperform image-based methods in both mIoU and mAcc. In addition, we propose a new metric that captures room integrity and boundary regularity, which confirms that our method produces much more regular segmentations. Source code is available at <https://github.com/DrZiji/VecFloorSeg>.

1. Introduction

Vector graphics are widely used in industrial designs, including graphic designs [26], 2D interfaces [5] and floorplans [15]. In particular, 2D floorplans consisting of geometric primitives (e.g., lines and curves) are the de-facto

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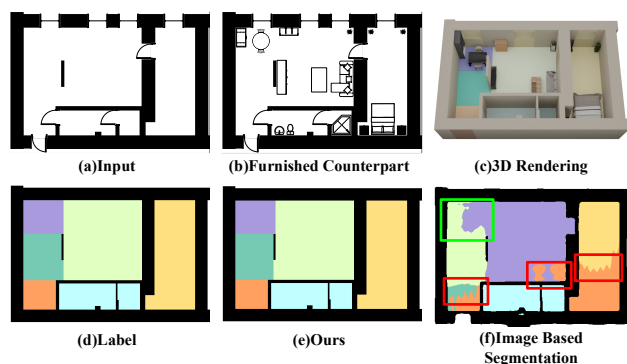


Figure 1. Comparing the results of our vector graphics based method (e) and raster image-based method [39] (f). Our result has straight boundaries and consistent region labels, compared with image-based result where red squares highlight semantic confusion and the green square underscores missing room prediction.

data representation for interior designs, indoor construction and property development. In contrast to raster images with fixed resolutions, vector graphics can be arbitrarily scaled without artifacts such as blurring and aliasing details. On the other hand, due to the irregularly structured data, it is difficult to apply image-based backbone networks directly to vector graphics for various applications.

Semantic segmentation of roughcast floorplans into rooms with labeled types (e.g. bedroom, kitchen, etc.) is a fundamental task for downstream applications. Interior designers usually first draw the roughcast floorplan, including basic elements like wall blocks and pipe barrels for property development (Fig. 1(a)) [29, 34]. Afterwards, interior furnishing, furniture layout, and 3D room spaces can be constructed and customized (Fig. 1(b)&(c)) [33]. During this procedure, it is important to obtain semantic segmentation of room spaces to cater to above needs. While recognizing room layouts from wall structures is straightforward for humans, automatic recognition with accurate semantics and clean boundaries is challenging.

Recent works [19, 22, 23, 39] use powerful image-based segmentation networks on rasterized floorplans to predict room segmentation in a pixel-wise manner. Due to the pixel-wise processing that ignores the integrity of structural elements, their results tend to have jigsaw boundaries and fragmented semantic regions as shown in Fig. 1(f). Besides, these methods usually rely on texts and furniture to determine the semantic labels, which are not available in roughcast floorplans. Another line of prior works processes vector graphics for recognition, e.g., object detection [14, 28] and symbol spotting [9, 10, 41]. However, to our best knowledge, semantic segmentation of vector graphics, roughcast floorplans in particular, has not been investigated before.

In this work, we make a first attempt at semantic segmentation of 2D roughcast floorplans directly as vector graphics. On one hand, by working with vector floorplans directly, the segmentation output is naturally regular and compact vector graphics rather than dense pixels (cf. Fig. 1(e)&(f)), which greatly facilitates downstream applications. On the other hand, the vector roughcast floorplans pose challenges in the following aspects. First, rooms in vector floorplans seldomly contain complete contour lines formed by the input line segments (see Fig. 1(a)&(d)). Second, the type of a room is determined not only by its shape but also by the relative relationships with its neighboring rooms and within the overall floorplan.

To address the above challenges, we make two observations. First, room spaces can be subdivided into a set of polygonal regions by input lines together with their extensions (Fig. 2), and their semantic classification as room types defines room segmentation. Second, lines (including extended lines) are potential boundaries of different rooms, and their being classified as boundaries or not should assist room segmentation in a dual direction.

Based on the two observations, we design a two-stream graph attention network (GAT) for the task. As illustrated in Fig. 2, the primal stream takes as input the primal graph that encodes line endpoints as vertices and line segments as edges, and predicts the boundary classification of edges; the dual stream takes as input the dual graph that encodes partitioned regions as vertices and their adjacency as edges, and predicts the vertex classification of regions, which effectively defines the semantic segmentation of a vector floorplan. Furthermore, the two streams should enhance each other rather than being separated. To facilitate data exchange between two streams, we present a novel modulated GAT layer to fuse information from one stream into the graph network computation of the other stream. We evaluate our approach on two large-scale floorplan datasets; both classical metrics and a new metric that we develop to focus on integral segmentation show that our results improve previous image-based results significantly. To summarize, we make the following contributions:

- We approach semantic segmentation of vector roughcast floorplans through the dual aspects of boundary line classification and region classification.
- We design two-stream graph neural networks to process dual regions and primal lines respectively, and devise a novel modulated GAT layer to exchange data across streams.
- We propose a new metric to capture both accuracy and integrity of the segmentation results.
- We obtain vector segmentation results on two floorplan datasets, which show much more compact boundaries and better integrity than raster image-based results.

2. Related Work

Rasterized Floorplan Understanding Floorplan understanding plays a crucial role in various indoor applications, such as interior designs, room space modeling, and real estates. Traditional methods [1, 8] treat a floorplan as an image and rely on low-level image processing and pre-defined rules to detect walls, doors and windows. However, they are limited by hand-crafted features and cannot handle floorplan elements with diverse drawing styles, e.g., different shapes, lengths or thickness. Recently, deep learning-based floorplan segmentation dominates the field and can achieve superior performance. For example, previous works [15, 19] employ a fully convolutional network (FCN) [20] to recognize room types, boundaries, and indoor furniture. Zeng et al. [39] design a deep multi-task network to learn room types and boundaries, respectively, and model their spatial interactions by boundary-guided attentions. Latest works [22, 23, 30] employ more powerful semantic segmentation networks [6, 27] and utilise object detection to detect furniture and texts, which as semantic priors further enhances the segmentation performance. However, image-based room segmentation is easily interfered by pixel noises and drawing styles, leading to aliased predictions near room boundaries and outlier fragments in segmented rooms. To avoid these drawbacks, this work explores direct segmentation with vectorized floorplans. After extending the lines in a floorplan to partition space into regions, we tackle the vector floorplan segmentation task with two complementary tasks, i.e. line classification as room boundaries and region classification as room types; by operating on these regular elements, the integrity of result segments largely improves.

Vector Graphics Recognition Vector graphics are widely used in 2D CAD designs, urban designs, graphic designs, and circuit designs, to facilitate resolution-free high precision geometric modeling. Considering their wide applications and great importance, many works are devoted to

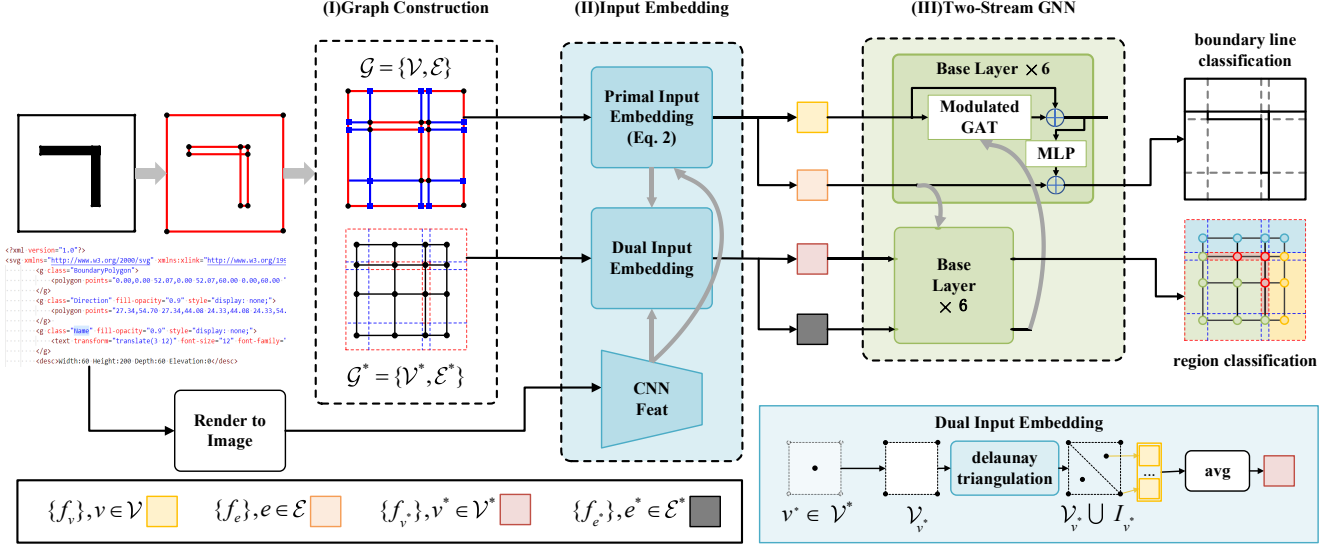


Figure 2. The overall pipeline of the proposed method. Given a vector floorplan input, we first build the primal and dual graphs in stage I, then compute the vertex/edge embeddings for the graphs in stage II by referring to CNN image features extracted from a rendered image of the floorplan, and finally in stage III use two parallel streams of GNN with modulated GAT layers exchanging data among them to learn the two tasks: primal edge classification as boundaries and dual region classification as room types. Since the two GNN streams have identical architecture, only one path has the layer operations shown. Lower right inset shows the feature embedding scheme for dual vertices.

recognition tasks on vector graphics. Fan et al. [10] propose panoptic symbol spotting on CAD floorplans to recognize different symbols. In subsequent works [9, 41], better network designs are explored to improve the spotting performance. Jiang et al. [14] explore vectorized object detection and achieve a superior accuracy to detection methods [3, 18] working on raster graphics, while enjoying faster inference time and less training parameters. Shi et al. [28] propose a unified vector graphics recognition framework that leverages the merits of both vector graphics and raster graphics. In this work, we tackle vectorized floorplan segmentation, which differs from symbol recognition in that the semantic regions generally have no corresponding entities in the input. We construct primal extended line graphs and dual partitioned region graphs to enable segmentation learning.

Graph Neural Networks Pioneer works on Graph Neural Networks (GNN) [12] extend traditional neural networks to graph structured data, e.g., social networks [13] and recommendation systems [37]. More variants of GNNs, such as graph convolutional network (GCN) [16], graph attention network (GAT) [32], and other modifications [2, 4, 17], have been proposed and demonstrated ground-breaking performances for different graph learning tasks. Among these models, GAT features simplicity and generality, as it learns adaptive attention weights for neighbors by considering their similarity and relevance to the center node [4, 32, 40]. In this work, we design a novel graph attention module, to ensure the heterogeneous information from two different streams can modulate the attention operations of their dual streams, which enables the joint enhancement of the

two tasks with floorplan segmentation.

3. Methodology

This work aims at semantic segmentation of 2D vector roughcast floorplans. Being roughcast floorplans, the input consists of predominantly a set of 2D geometric line segments representing wall structures, along with few circular arcs representing other structures, and the output is the vectorized semantic segmentation of different room regions. The overall pipeline of our proposed method is illustrated in Fig. 2. First, we divide the whole floorplan into non-overlapping polygonal regions by extending the line segments until intersection. Then, we construct two graphs to represent the primal geometric line segments and the dual polygonal regions, and calculate their feature embeddings as described in Sec. 3.1 and Sec. 3.2. Finally, we use a two-stream GNN for segmentation, where the primal stream predicts whether or not a line segment separates different room regions, and the dual stream classifies the partitioned polygon regions into different room types (see Sec. 3.3), with two streams enhancing each other. In this way, semantic segmentation of a vector floorplan is reformulated as polygon region classification, which naturally avoids imprecision in rasterization based methods.

3.1. Graph Construction

Aesthetically pleasing and rational room layouts contain both rooms that are separated into private areas, for example, bedrooms, and rooms that are connected without wall separation, e.g. living rooms, corridors, dining rooms, etc.

The partition of rooms without explicit separating walls is ambiguous. From the perspective of floorplan designers, a good room region is usually as regular and rectangular as possible; thus interior designers usually partition room space with extended wall structures as illustrated in Fig. 2(I). In this work, we follow this intuition and heuristically extend line segments so that the design space is divided into a set of polygonal regions (details about region partition are given in the supplementary). In this way, we can avoid rasterization and achieve semantic segmentation directly by classifying the room type of each partitioned region.

To learn the segmentation based on the spatial relation between wall structures and region layouts, we construct graphs to encode it. We represent the input capturing line structures with a primal graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ denotes the set of end points of line segments, and $\mathcal{E}$ denotes the set of line segments as shown in Fig. 2 (I). A vertex $v_k \in \mathcal{V}$ has 2D coordinates $\mathbf{v}_k \in \mathbb{R}^2$, while an edge $e_{ij} \in \mathcal{E}$ connecting vertex v_i and v_j is encoded by the tuple $(v_i, v_j, \cos(\theta_{e_{ij}}), c_{e_{ij}})$, where $\theta_{e_{ij}} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ is the angle between e_{ij} and the x-axis and $c_{e_{ij}} \in \{0, 1\}$ indicates whether e_{ij} is an extended line segment or an original line segment in vector floorplan. Note that c_e is significant, as a line segment from the input is highly likely to represent a wall and room boundary, while an extended line segment is only a possible room boundary.

We capture the partitioned polygonal room regions with a dual graph where the regions are represented by nodes and their adjacencies are defined by edges. The dual graph is defined as $\mathcal{G}^* = \{\mathcal{V}^*, \mathcal{E}^*\}$ as demonstrated in Fig. 2(I). A vertex $v^* \in \mathcal{V}^*$ in the dual graph denotes the polygonal region, which corresponds to a sub-graph $\mathcal{G}_{v^*} = \{\mathcal{V}_{v^*}, \mathcal{E}_{v^*}\} \subset \mathcal{G}$ from the primal graph that bounds the region. For simplicity, we take the mean value of primal vertex attributes (i.e. 2D coordinates) of $\mathcal{G}_{v^*}$ as the attribute of the dual vertex v^* . An edge $e_{ij}^* \in \mathcal{E}^*$ in the dual graph indicates the adjacency of two polygonal regions and we obtain its attribute as the concatenation of attributes of its two endpoints (v_i^*, v_j^*) .

We note there is a correspondence between primal and dual edges as well. If $v_i^*, v_j^* \in \mathcal{V}^*$ has an edge e_{ij}^* , their corresponding polygonal regions $\mathcal{G}_{v_i^*} = \{\mathcal{V}_{v_i^*}, \mathcal{E}_{v_i^*}\}$ and $\mathcal{G}_{v_j^*} = \{\mathcal{V}_{v_j^*}, \mathcal{E}_{v_j^*}\}$ would share primal line segment expressed as $\mathcal{E}_{v_i^*} \cap \mathcal{E}_{v_j^*}$. On the other hand, without loss of generality, we assume a primal edge $e \in \mathcal{E}_{v_i^*} \cap \mathcal{E}_{v_j^*}$ is adjacent to regions v_i^*, v_j^* , and it has a corresponding dual edge, which is exactly e_{ij}^* . To summarize, we define the dualization operation $*$ on edges by

$$\begin{aligned} *(e_{ij}^*) &= \mathcal{E}_{v_i^*} \cap \mathcal{E}_{v_j^*} \\ *e &= e_{ij}^*, \quad e \in *(e_{ij}^*) \end{aligned} \quad (1)$$

Theoretically speaking, $\mathcal{E}_{v_i^*} \cap \mathcal{E}_{v_j^*}$ only contains a single primal edge. We keep the set expression in Eq. 1 since there

may be extra vertices separating a straight wall into multiple primal edges. Please refer to the supplementary for an illustration of such exceptions.

3.2. Input Embedding

After constructing two graphs, we embed vertex and edge attributes as input features to the two-stream GNN. In addition, despite the equivalence of raster images and vector floorplans in depicting room layout, we find that using features obtained from the raster image as additional input helps perceive larger spatial receptive field and improves the performance (Sec. 4.5); thus we use both image features and geometric features for vertex embedding.

The embedding for primal graph vertices $\mathcal{V}$ is given by

$$f_v = x_v + pe_v, \quad v \in \mathcal{V} \quad (2)$$

where $x_v \in \mathbb{R}^C$ denotes the feature vector indexed by the coordinates of v from the rasterized floorplan image feature, and $pe_v \in \mathbb{R}^C$ is the sinusoidal positional encoding of vertex positions following [31]; please refer to supplementary for more details.

The embedding of a dual vertex $v^* \in \mathcal{V}^*$ should capture the features of a polygonal region. Thus we compute it as the average feature of points sampled from the polygonal region enclosed by $\mathcal{G}_{v^*} = \{\mathcal{V}_{v^*}, \mathcal{E}_{v^*}\}$. First, $\mathcal{V}_{v^*}$ is selected to maintain the corner information. Also, we sample interior points I_{v^*} by using the centers of Delaunay triangles [25] within $\mathcal{G}_{v^*}$. The features of I_{v^*} are calculated in the same form as Eq. 2. The dual vertex embedding feature f_{v^*} is calculated as

$$f_{v^*} = \frac{1}{|\mathcal{V}_{v^*} \cup I_{v^*}|} \sum_{v \in \mathcal{V}_{v^*} \cup I_{v^*}} f_v, \quad v^* \in \mathcal{V}^* \quad (3)$$

The sampling strategy and average aggregation are illustrated in the lower right inset of Fig. 2 (Dual Input Embedding). We use this sampling strategy for its simplicity, any other strategies are also permitted (e.g. uniform sampling).

We embed the primal and dual edges by mapping their attributes to input features. Since a primal edge $e \in \mathcal{E}$ contains attributes other than endpoint positions (see Sec. 3.1), we map the coordinates and additional attributes to C -dim features by separate learnable projection matrices W_e, W_θ :

$$f_{e_{ij}} = W_e(pe_{v_i} || pe_{v_j}) + W_\theta(\cos(\theta_{e_{ij}}) || c_{e_{ij}}), e_{ij} \in \mathcal{E} \quad (4)$$

where $||$ denotes feature concatenation. For dual edges $\mathcal{E}^*$ which mostly depict the adjacency of dual vertices and lack additional attributes, we compute their feature embeddings simply from dual vertex coordinates:

$$f_{e_{ij}^*} = W_{e^*}(pe_{v_i^*} || pe_{v_j^*}), \quad e_{ij}^* \in \mathcal{E}^* \quad (5)$$

where $W_{e^*} \in \mathbb{R}^{C \times 2C}$ is a learnable projection matrix.

3.3. The Two-Stream Graph Neural Network

After embedding input features of two graphs, we feed them to a two-stream graph neural network to learn high-level semantic features. The two streams address complementary tasks, i.e., boundary line classification and polygonal region classification, which attacks the segmentation problem from dual aspects and should enhance each other.

For example, if two vertices in the dual graph $\mathcal{G}^*$ are assigned different labels, the shared line segments of their corresponding polygonal regions in $\mathcal{G}$ should be room boundary lines. Meanwhile, if an edge in the primal graph $\mathcal{G}$ is predicted as room boundary, regions that are separated by the edge are likely to have different room types. To facilitate such edge based modulation, we devise a novel modulated GAT layer that learns vertex features in one stream by taking edge features from the other stream into account.

As shown in Fig. 2, the two-stream graph neural network takes feature embeddings of the primal and dual graphs as input and applies two parallel graph neural networks, each with $L = 6$ modulated GAT layers. For the l th-level modulated GAT layer the vertex feature $f_{v_i}^l$ in the primal (dual) stream accumulates features from its neighboring vertices N_{v_i} as follows:

$$f_{v_i}^{l+1} = f_{v_i}^l + \Theta_\gamma(\alpha_{ii}W_v f_{v_i}^l + \sum_{v_j \in N_{v_i}} \alpha_{ij}W_v f_{v_j}^l) \quad (6)$$

where Θ_γ denotes an MLP layer, $W_v \in \mathbb{R}^{C \times C}$ is a learnable projection matrix, and α_{ij} is an adaptive weight to balance neighborhood features.

We compute α_{ij} as:

$$\alpha_{ij} = \frac{\exp([W_q f_{v_i}^l]^T \cdot g(f_{*e_{ij}}^l) \cdot [W_k f_{v_j}^l])}{\sum_{v_m \in N_{v_i} \cup \{v_i\}} \exp([W_q f_{v_i}^l]^T \cdot g(f_{*e_{im}}^l) \cdot [W_k f_{v_m}^l])} \quad (7)$$

where $W_q, W_k \in \mathbb{R}^{C \times C}$ are learnable parameters and $g(\cdot) : \mathbb{R}^C \rightarrow \mathbb{R}^{C \times C}$ maps the *dual edge feature* from the other stream to a weight matrix, which we use to modulate the similarity between two vertices. For simplicity, we implement $g(f_{*e_{ij}}^l) = W_g f_{*e_{ij}}^l$, with a learnable tensor $W_g \in \mathbb{R}^{(C \times C) \times C}$. Moreover, for the degenerate case of e_{ii} , we simply set $g(f_{*e_{ii}}^l) = I$, where I is the identity matrix.

The dual edge features are obtained by the correspondence established in Eq. 1. Specifically, for a primal edge e , the dual edge $*e \in \mathcal{E}^*$ is unique and its feature is directly obtained from the dual stream GNN. For a dual edge e^* , its corresponding primal edges are represented by a set $*(e^*)$ and we obtain the feature as the average of all primal edges in this set.

After the feature aggregation on vertices, we update the edge features of two streams in the l th-level modulated GAT

layer by:

$$f_{e_{ij}}^{l+1} = f_{e_{ij}}^l + \Theta_e(f_{v_i}^l || f_{v_j}^l), \quad e_{ij} \in \mathcal{E} \cup \mathcal{E}^* \quad (8)$$

where Θ_e denotes an MLP layer.

The output features of the two-stream GNN are used for prediction tasks. The edge features in $\mathcal{G}$ produced by the primal stream are fed into a binary classification head to predict whether or not an edge in $\mathcal{G}$ separates different room regions. The vertex features in $\mathcal{G}^*$ from the dual stream are projected into the probabilities of the vertices belonging to different room types.

3.4. The Objective Function

We optimize the two-stream network by minimizing the boundary classification loss for lines $\mathcal{L}_p$ and room type classification loss for regions $\mathcal{L}_r$. $\mathcal{L}_p$ classifies primal line segments as lying on the boundaries of rooms by the binary cross entropy. $\mathcal{L}_r$ classifies dual polygonal regions into corresponding room types by the cross entropy, which also gives the semantic segmentation of vector floorplans. We employ the focal loss on $\mathcal{L}_r$ for better performance. The overall loss $\mathcal{L}$ is defined as $\mathcal{L} = \lambda_p \cdot \mathcal{L}_p + \mathcal{L}_r$, where the hyper-parameter λ_p is set to 0.5 empirically.

4. Experiments

4.1. Datasets

We conduct experiments on two datasets including R2V [19] and CubiCasa-5k [15]. R2V is a floorplan segmentation dataset covering 10 semantic categories (i.e., wall, door, window, washing room, bedroom, closet, balcony, hall, kitchen, other room, background) and consisting of 870 images, where 770 images are used as training data and the remaining 100 images are used for testing. We convert each image to its corresponding roughcast vector floorplan by extracting the contours of wall masks as closed line loops and converting line primitives into SVG commands.

CubiCasa-5k is a large-scale floorplan dataset with both vector graphics (SVG format) and rendered images, in which annotations of 12 room types (background included) and 11 furniture types are available. As we focus on roughcast floorplan segmentation, we only keep the walls as input and the room annotations as ground truth. We further remove data items with unlabeled rooms, thus having 4192 items for training, 399 for validation, and 400 for testing.

4.2. Implementation Details

Input We rescale the longer side of a raster floorplan to 256 and keep its aspect ratio, and pad the short side with white color, to form an input image with size of 256×256 for feature extraction. 2D coordinates of graph vertices are correspondingly uniformly scaled to the range $[0, 255]$.

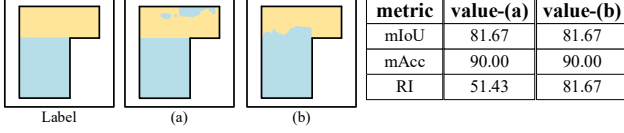


Figure 3. Illustration of our proposed RI metric for room prediction integrity.

Training Our implementation is based on prevalent GNN code base PyG [11]. We train the network with a batch size of 8. We use the SGD optimizer with momentum $\mu = 0.9$ and set weight decay to 0.0005. The CNN image feature backbone is loaded with pre-trained weights provided by mmsegmentation [7] and fine-tuned during training. Training data is randomly flipped with a probability of 0.5 as data augmentation. The network is trained for 200 epochs with a learning rate initialized as 0.01 and scheduled with the cosine annealing strategy [21].

4.3. Evaluation Metrics

Following existing works on floorplan segmentation [23, 39], we employ mean Intersection over Union (mIoU) and class-wise mean Accuracy (mAcc). However, these metrics count pixels and are insensitive to fragmented and noisy segments which severely affects the room integrity and boundary smoothness (Fig. 1). These artifacts, however, pose great difficulty for downstream applications like 3D modeling. The proposed RI metric is proposed to penalize fragmented segments. It accounts for room integrity by treating regions with consistent labels as rooms (see Fig. 3), so that the outlier fragments would be penalized by being exposed as independent rooms.

First, we build a match between the predicted rooms and ground truth rooms. Since the predicted rooms P and the ground truth rooms G are sets with possibly different cardinalities, we use the optimal bipartite matching to build correspondence [24], where the cost metric $C(p, g) = 1 - \text{IoU}(p, g)$ is defined for each pair of prediction $p \in P$ and ground truth $g \in G$. Given the bipartite correspondence $\sigma : P \rightarrow G \cup \{\phi\}$, we say a predicted room $p \in P$ is matched to a ground truth room $\sigma(p) \in G$ if $\text{IoU}(p, \sigma(p))$ is greater than 0.5 and p and $\sigma(p)$ have the same room type.

Formally, we define RI (ranged [0,1]) as the product of the average IoU of matched rooms and F1-score:

$$RI = \frac{\sum_{(p,g) \in TP} \text{IoU}(p, g)}{|TP|} \cdot 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (9)$$

$$= \frac{\sum_{(p,g) \in TP} \text{IoU}(p, g)}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|},$$

where TP are matched prediction and ground truth pairs, FP are predicted rooms without matched ground truth, and

Methods	Backbone	Params(M)	GFLOPs	mIoU	mAcc	RI
DFPR [39]	VGG-16	28.91	223.39	69.47	81.5	50.96
Ours	—	22.57	91.63	75.56	87.32	80.48
DeepLabV3+ [6]	ResNet-50	43.59	176.76	74.59	83.46	49.99
DNL [36]	—	50.02	200.16	71.61	81.90	51.99
UperNet [35]	—	66.41	237.02	73.23	83.84	63.46
Ours	—	33.29	115.15	79.77	88.41	84.67
OCRNet [38]	ResNet-101	55.51	231.11	77.98	85.34	70.82
Ours	—	51.39	193.01	81.38	89.86	86.20

Table 1. Comparison results on R2V.

Methods	Backbone	val-set			test-set		
		mIoU	mAcc	RI	mIoU	mAcc	RI
DFPR [39]	VGG-16	49.68	60.37	38.44	47.73	58.68	38.57
Ours	—	60.27	72.32	66.89	57.48	69.89	64.09
DeepLabV3+ [6]	ResNet-50	60.46	73.18	38.41	58.18	71.75	35.16
DNL [36]	—	59.61	72.15	42.38	55.29	68.36	40.49
UperNet [35]	—	59.31	72.22	45.90	57.04	70.50	44.71
Ours	—	63.09	75.48	69.74	61.35	74.45	67.99
OCRNet [38]	ResNet-101	60.44	72.94	43.99	57.13	70.62	41.89
Ours	—	64.36	76.98	69.55	62.49	75.48	67.51

Table 2. Comparison results on CubiCasa-5k.

FN are ground truth without matched predicted rooms. The average IoU measures the segmentation quality of matched pairs, while the F1-score evaluates segmentation integrity by counting matched/unmatched regions rather than pixels. To this end, we show in Fig. 3 that results with fragmented regions harmful to the room integrity are much more penalized in RI metrics.

4.4. Comparison

We compare our method with prevalent image-based semantic segmentation methods, e.g., DeepLabV3+ [6] and OCRNet [38], which are also widely used in floorplan segmentation, as well as DFPR [39] that specializes for floorplans. We report the results in Tab. 1 & 2. The quantitative results show that our proposed method based on vector floorplans outperforms all other baselines in terms of mIoU, mAcc and room integrity (RI) under different image backbones, and shows remarkable strength on RI, despite that our network has the least amount of parameters. The reason for the poor RI performance of baselines is that they tend to produce many fractured regions with noisy semantics as shown in Fig. 4. Under RI, the outlier patches on one hand become false positives and yield a larger denominator in Eq. 9, and on the other hand prevent correct matching pairs and increase false negatives, all contributing to lower RI.

The qualitative comparison is provided in Fig. 4, the color bar of room categories is displayed in the supplementary. Although DeepLabV3+ and OCRNet (column (d)&(e)) expand receptive fields and integrate contextual features to improve the performance, their outputs still suffer from aliasing boundaries and inconsistent segmentation. DFPR integrates boundary information to assist room segmentation, but its performance on roughcast floorplans is

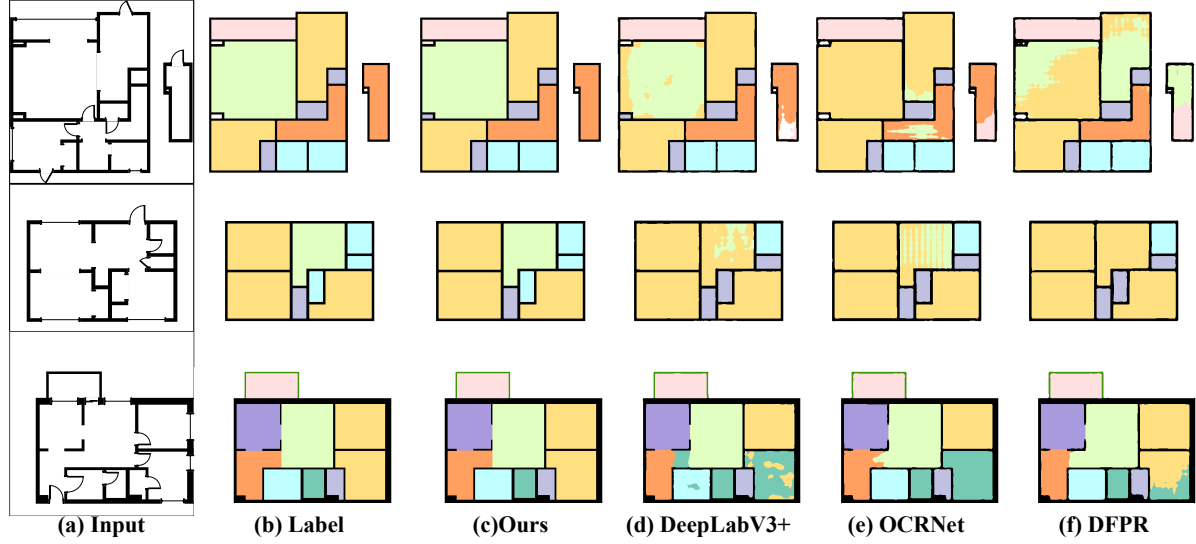


Figure 4. The qualitative comparison with image-based floorplan segmentation methods on two datasets. The first two rows are examples from R2V and the last row is from Cubicasa.

Vertex Embedding			Architecture		mIoU	mAcc	RI
pos.	vertex samp.	backbone	p-stream	GAT Ours			
✓					45.92	60.30	53.45
✓					59.01	73.29	65.12
✓	✓				61.32	75.03	67.35
✓	✓	✓			75.60	85.34	81.03
✓	✓	✓	✓		77.08	85.77	83.04
✓	✓	✓	✓	✓	79.77	88.41	84.67

Table 3. Ablation studies on vertex embedding and the network architecture. *pos.*: using sine positional encodings in Eq. 2. *vertex samp.*: using interior sampling (Eq. 3) to compute the dual vertex embedding. *backbone*: adding image features from rasterized floorplans. *p-stream*: with the primal stream that predicts boundary lines. *GAT Ours*: using the modulated GAT layer that enables feature interaction between two streams; if not marked, the GAT layer degrades to vanilla similarity-based attention [31] by replacing the learnt weight matrix $g(f_{*e_{ij}}^t)$ in Eq. 7 with an identity matrix. If none of the above is marked, pe_v of Eq. 2 is set to zero, and x_v is a learnable embedding indexed by vertex category (i.e. primal vertex or sampled vertex). $W_e(\cdot)$ and $W_{e*}(\cdot)$ of Eqs. (4,5) are also set to zero.

greatly degraded as no interior textures or text symbols are available for shortcut semantic inference. When zoomed in, the predicted boundaries show serious sawteeth patterns, which would impose additional burdens on post-processing and down-stream applications. By contrast, our method takes advantage of the regular line segments and regions to generate smooth boundaries and consistent room segmentation.

4.5. Ablation Studies

In this part, we analyze the effectiveness of vertex embedding, architectural designs of the two-stream GNN, as well as edge features. We use ResNet-50 as the backbone

dual map	edge ini.	mIoU	mAcc	RI
✓		77.38	86.37	83.95
✓		78.22	87.67	83.06
✓	✓	79.77	88.41	84.67

Table 4. Ablation studies on edge features. *dual map*: using dual edge features from the other stream (cf. Fig. 2(III)); if not marked, the modulated GAT in each stream uses edge features from its own stream instead. *edge ini.*: using the edge embeddings of Eq. 4 and 5; if not marked, the edge embeddings are set to zero.

in all ablation studies.

Vertex Embedding for primal and dual graphs involves positional encodings and image features (Sec. 3.2). Results in Tab. 3 upper part reveal their significance for good performance. In particular, an alternative to the embeddings of dual vertices in Eq. 3 is to consider the region centroid only instead of interior sampling, and results show the sampling strategy works better.

The Two-Stream GNN consists of a necessary dual stream for segmentation prediction, and a primal stream for boundary line classification that can be removed, with their interactions implemented by our proposed modulated GAT layer. Results in Tab. 3 lower part suggest that the inclusion

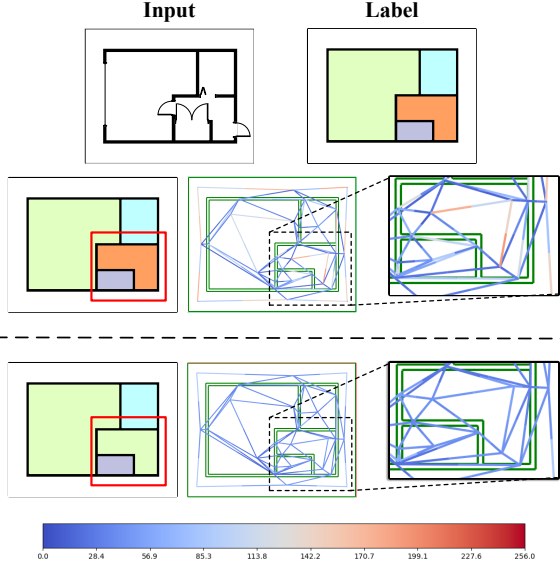


Figure 5. Visualization of attention weights generated by the modulated GAT (second row) and vanilla similarity-based attention [31] (third row), where from left to right are segmentation results, the attention visualization and their zoom-ins. For attention visualization, the dual graph $\mathcal{G}^*$ is superimposed on room boundaries (green lines); an edge $e_{ij}^* \in \mathcal{E}^*$ is colored by corresponding attention weights α_{ij}^* and α_{ji}^* . Since $\alpha_{ij}^* \neq \alpha_{ji}^*$ are asymmetric, we split the edge at the middle and draw two parts with colors encoding α_{ij}^* and α_{ji}^* . Warmer color means larger attention. Best viewed by zooming in.

of the primal stream and the modulated GAT layer both improve the performance significantly (4.17% in mIoU, 2.97% in mAcc and 3.64% in room integrity), which means that on one hand the boundary classification task enhances segmentation through their shared embedding layers, and on the other hand the modulated GAT layer facilitates such mutual enhancement further.

Edge Features are a critical detail for the modulated GAT layer communicating between two GNN streams. First, we find the edge embedding (Eqs. 4 and 5) contributes to the boundary prediction in the primal stream and room segmentation in the dual stream, because by fixing other network settings and only setting the edge embedding to zero, the final segmentation results shown in the second row of Tab. 4 degrade. On top of that, we find the modulation by dual edges in GAT layers is important too, because by switching the modulation in primal(dual) stream to edge features of their streams, the results also degrade as shown in the first row of Tab. 4.

4.6. Discussion

Visual Analysis on the Modulated GAT To demonstrate how feature interaction between two-stream GNN works, we visualize the attention weights in the last GNN layer of the dual stream in Fig. 5. The visualization results show that

the attention weights can capture clear region and boundary patterns under dual edge feature modulation. For example, the attention weights are small if two connected vertices lie in different rooms, confirming that the feature interaction between vertices is suppressed; contrarily, the attention weights become greater when they are in the same room. In comparison, GAT layers with vanilla attention (i.e. replacing $g(f_{*e_{ij}}^l)$ in Eq. 7 with an identity matrix) do not show such obvious adaptiveness, which explains the performance difference in Tab. 3 lower part.

Limitations Despite producing superior results compared with image-based floorplan segmentation methods, our method cannot give accurate predictions in several scenarios. For example, for extremely small regions within a floorplan, our method tends to assign wrong labels, partly because these regions have few meaningful interior samples to produce good vertex embedding features. Another case is the misclassification of similar categories (e.g. wall blocks and railings), which we may leverage more detailed shape priors like door and window features to tackle in the future. Besides, our method can only handle line primitives and use polylines as a substitution for curves at present, which makes it difficult to process complex curves, e.g., B-Spline and Bezier curves. Illustrations of discussed limitations can be found in the supplementary.

5. Conclusion

In this work we present a novel semantic segmentation problem on a typical kind of vector graphics, i.e., to process vector roughcast floorplans directly and produce compact and regular segmentation. While existing image-based segmentation methods are not directly applicable, we propose a two-stream graph attention network directly working on vector graphics and cast the problem into dual tasks of room boundary classification and partitioned region classification. A novel modulated GAT module is devised to enable efficient interactions between the two streams, and therefore their mutual enhancement. Results and new metrics show our method achieves superior performance and produces much more regular and integral floorplan segmentation. Considering the importance of vector graphics for industrial design, we hope this work can inspire future research on vector graphics based deep learning for intelligent industrial design and analysis.

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